

COURSE: CS 250 — Theory of Computation

Credits: 3 | Department: Computer Science

Document ID: CAT-COURSE-CS250

Source: <https://catalog.techstate.edu/courses/cs250>

Date Accessed: March 29, 2025

PREREQUISITES: CS 201 AND MATH 231

CO-REQUISITES: None

OFFERED: Fall, Spring

DESCRIPTION:

Formal models of computation and their limitations. Topics include finite automata and regular languages, context-free grammars and pushdown automata, Turing machines, decidability and undecidability (the Halting Problem), reducibility, and complexity theory introduction (P, NP, NP-completeness). Emphasis on mathematical proof-writing.

GRADING: Problem sets 50%, Midterm 25%, Final 25%.